

The empirical recurrence for OEIS A253988, recovered from its tail

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Abstract

OEIS A253988 records “Empirical recurrence of order 92” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 701$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 8$, so the recurrence is proved for every $n \geq 100$. Its last coefficient is $c_{92} = 8916100448256 \neq 0$, so no further tail satisfies anything shorter; any order-92 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A253988 is “Number of $(n+2) \times (3+2)$ 0..1 arrays with every 3×3 subblock diagonal median plus antidiagonal median nondecreasing horizontally, vertically and ne-to-sw antidiagonally.”. It has offset 1 and begins

11872, 68176, 477856, 3678188, 25361960, 175815629, 1239374238, 8586595803, ...

The entry states:

Empirical recurrence of order 92 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 14:23:56, revision 6) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 701$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 701$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 92: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 8$ is the first whose minimal order is exactly the stated 92.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 1402$ exact terms from $n = 8$ on, it returns order 92 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{92} c_i a(n-i),$$

c_1	14	c_{24}	−15865995639	c_{47}	109196822458174	c_{70}	39943335358447105
c_2	−48	c_{25}	−103862826720	c_{48}	4207869863849319	c_{71}	−2908632779272152
c_3	−56	c_{26}	−874673650731	c_{49}	−11245597131267548	c_{72}	−16360666429874856
c_4	543	c_{27}	−623791788604	c_{50}	5525255258046617	c_{73}	24699762504421056
c_5	−2922	c_{28}	2806133704621	c_{51}	−10294916148584262	c_{74}	−19195099753249872
c_6	12052	c_{29}	4171396587450	c_{52}	−2646561072913533	c_{75}	2965407883784064
c_7	−16346	c_{30}	−96466655344	c_{53}	24352218774814576	c_{76}	5492060032781952
c_8	−1342	c_{31}	1017320719444	c_{54}	−27524368018903709	c_{77}	−10706047928921088
c_9	429250	c_{32}	−9144541568548	c_{55}	40657194707001280	c_{78}	9019215548460288
c_{10}	−1975357	c_{33}	−17628577119372	c_{56}	−16772561740932002	c_{79}	659097927936000
c_{11}	1723510	c_{34}	−37100548242274	c_{57}	−15289123958027408	c_{80}	−5396384457381888
c_{12}	−1449032	c_{35}	−44668699369370	c_{58}	37722984746165848	c_{81}	4960837889998848
c_{13}	−14011898	c_{36}	163110487120734	c_{59}	−76600187161782448	c_{82}	−2825111573913600
c_{14}	101250927	c_{37}	189423573931502	c_{60}	61234157734045035	c_{83}	−46481737973760
c_{15}	−165328034	c_{38}	−50234822003049	c_{61}	−29089440499160942	c_{84}	1313557317844992
c_{16}	391710284	c_{39}	−77543706297772	c_{62}	−11636221863284096	c_{85}	−1661640185217024
c_{17}	980963266	c_{40}	9384174118171	c_{63}	69730366044243240	c_{86}	1298030463811584
c_{18}	−6319273167	c_{41}	−462406369696490	c_{64}	−76778009397922954	c_{87}	60513903968256
c_{19}	−1744082458	c_{42}	−840638484789850	c_{65}	66025472988660080	c_{88}	−696838656098304
c_{20}	11761127415	c_{43}	12794959853986	c_{66}	−31270135934668710	c_{89}	277389791723520
c_{21}	12585302152	c_{44}	99251881797108	c_{67}	−24167524937849434	c_{90}	58016570277888
c_{22}	36424624593	c_{45}	3216151010700178	c_{68}	46129056285589669	c_{91}	−54982619430912
c_{23}	−26079912346	c_{46}	−186986010487095	c_{69}	−55253452971280480	c_{92}	8916100448256

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 100$, and it is the recurrence OEIS A253988 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 8$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{92} - c_1x^{92-1} - \dots - c_{92}$. Its constant term is $-c_{92} = -8916100448256$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k\lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 92 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 92, so $\deg q = 92$, and it is monic. It holds from some index t on. From $\max(t, 8)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 92, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 17 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 92, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 8916100448256.

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-92 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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